The Culture of Business, Finance and the Economy

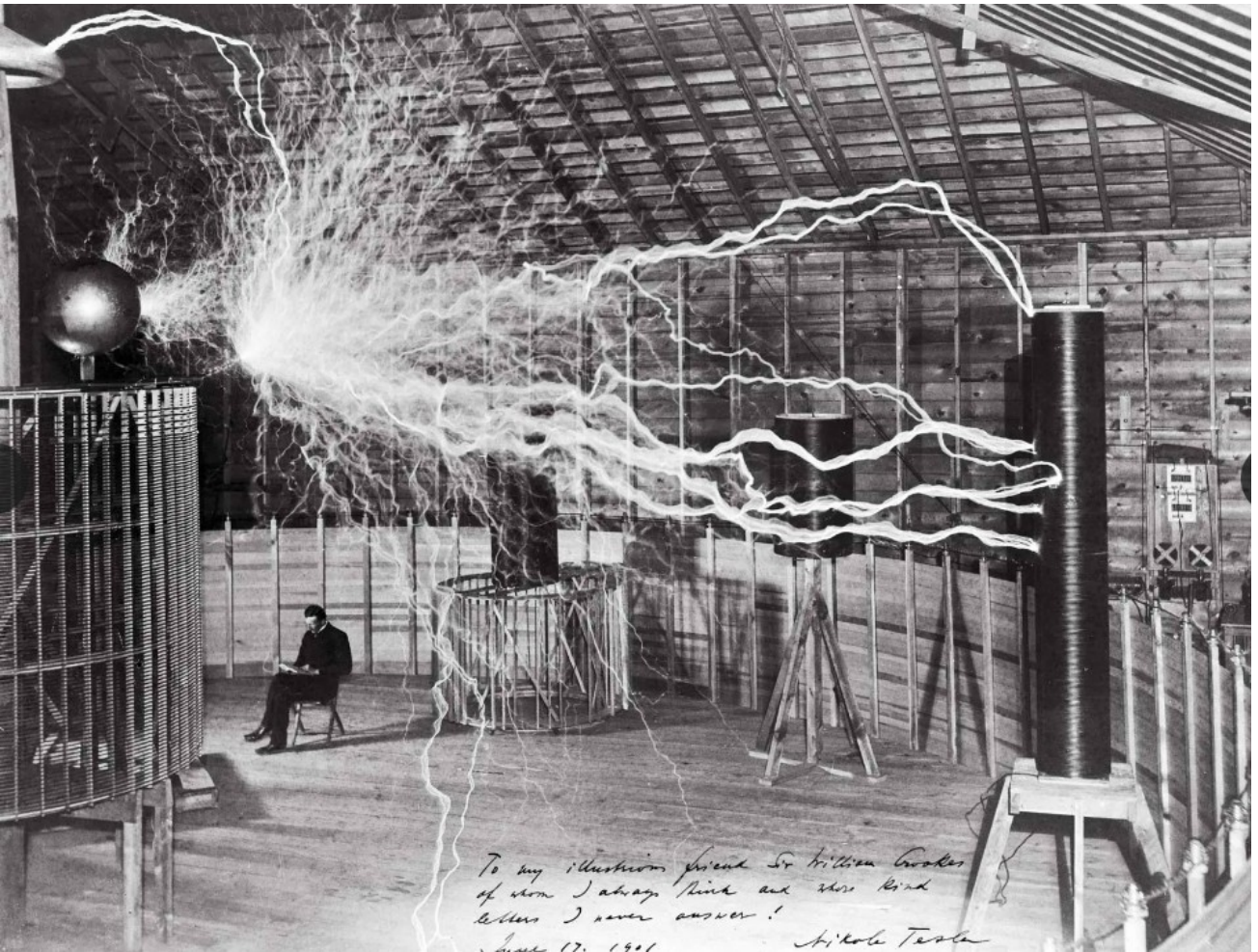
Class 6 – march 19

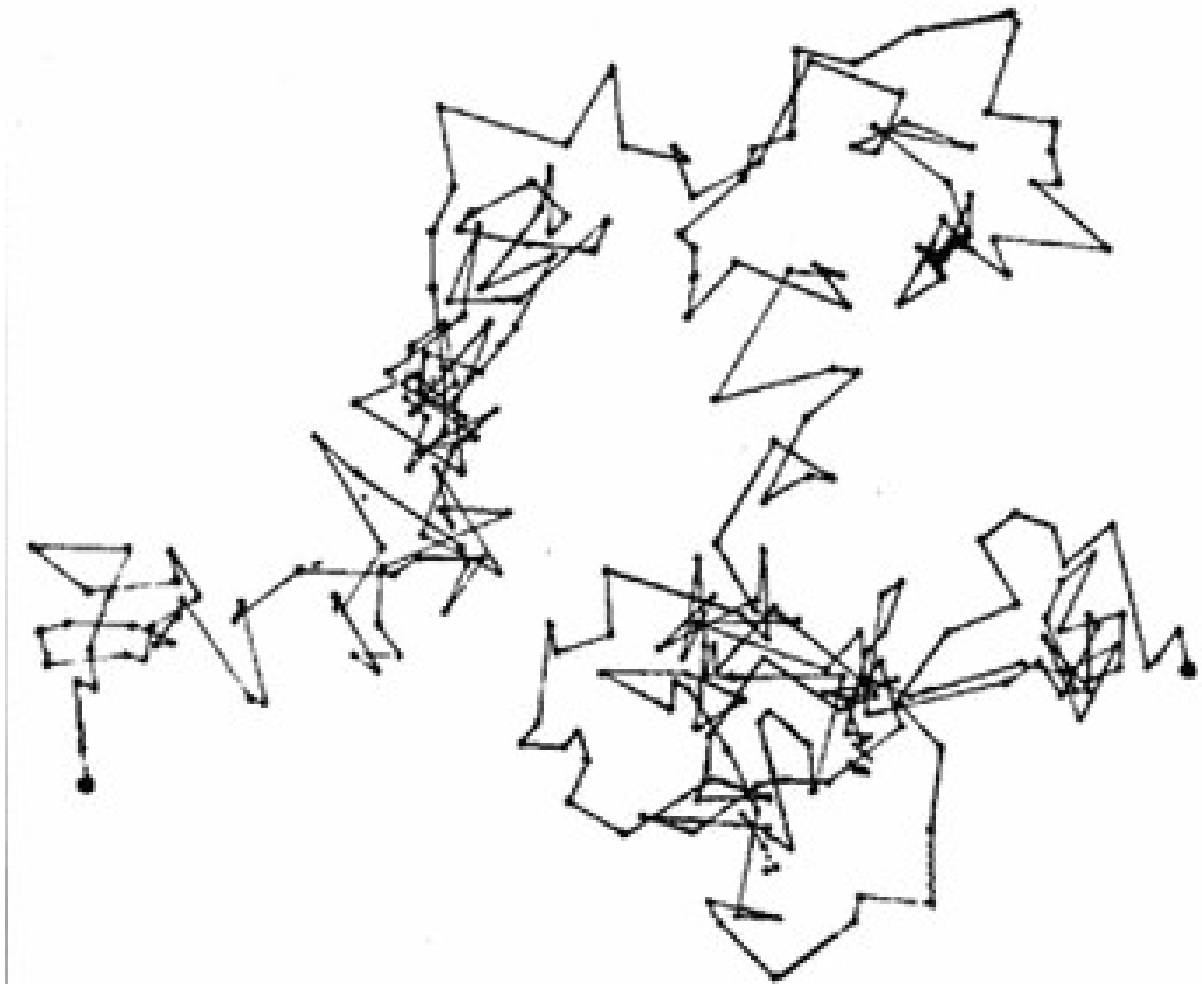
Dr. Juan M. del Nido

First semester 2026

Today

Economics and Physics: the science of all things





Recap

Mathematics and Economics

No Economics as we know it without Mathematics

Financial Markets: Negating the Laws of Gravity

Posted by Tothetick

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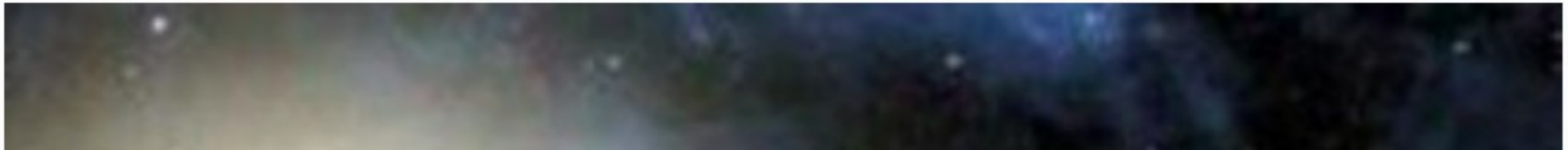
Forbes MÉXICO

Ethan Michaly
junio 17, 2014 @ 7:00 am

Desafiando las leyes de la gravedad



La Voz 120 AÑOS

Desafiar o negar la ley de la gravedad

El análisis económico del día. Es cierto que las políticas sociales son necesarias. No lo es que siempre convenga aumentar el gasto público. Hacerlo en contextos de recesión sin inflación es como desafiar la ley de la gravedad. Hacerlo cuando la inflación es alta es como negarla.

16 de mayo de 2014, 06:15

Gastón Utrera

HAL
open science

The Gravity Equation in International Trade: an Explanation

Thomas Chaney

Why is the world's financial plumbing under pressure?

3 December 2019

Tim Harford
Presenter, 50 Things That Made the Modern Economy

federalreserve.gov/econres/rsfof-staff.htm

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Flow of Funds

The Flow of Funds section produces the Board's quarterly Financial Accounts of the United States, a comprehensive set of accounts that includes detail on the assets and liabilities of households, businesses, governments, and financial institutions. Staff members analyze the effect of financial innovations and other changes in financial markets on the structure of the accounts. They also conduct research on issues concerning the relationships between economic sectors and financial markets and institutions, and they study ways to improve techniques for seasonal adjustment and estimation of missing data.

Floating the Moroccan Dirham: Challenges and Opportunities in 2026

Morocco is on the brink of a transformative economic reform as it prepares to transition to a floating exchange rate for the dirham by 2026. This historic move represents a strategic effort by the government to enhance the nation's economic resilience, attract foreign investment, and integrate more deeply into global financial markets. Designed to unlock long-term growth, this reform brings with it immediate risks that must be navigated with precision, requiring robust planning and economic stability.

by **Badr Bouarich** — Dec, 29, 2024

0 0 AA

Fixed Income

Investors increasingly worried over swelling US debt burden

US Treasuries came under heavy selling pressure this week, with investors increasingly worried over the country's swelling debt burden. House Republicans have quickly advanced President Donald Trump's sweeping tax and spending cut bill, which was then passed in the lower chamber of Congress.

by UBS Editorial Team | 23 May 2025

ADVERTORIAL

Market Volatility Impacts Pepe Investors as ETH Holders Explore Alternatives Like Remittix

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POLITICS

Defense lawyers for needy clients feel squeeze after congressional funding dries up

By **Melissa Quinn**
Updated on: August 11, 2025 / 11:24 AM EDT / CBS News

Washington – Just over a year ago, federal judges, defenders and a group of lawyers who represent criminal indigent defendants gathered at the U.S. courthouse in Baltimore to hear federal Judge Myron Thompson give an address.

THE ASTANA TIMES
BRINGING KAZAKHSTAN TO THE WORLD

Kazakh National Currency Weakens as Market Forces Drive Decline

BY AIMAN NAKISPEKOVA *in* EDITOR'S PICKS, NATION *on* 28 JULY 2025

ASTANA – Kazakhstan is experiencing a sharp depreciation of its national currency, as the exchange rate for the U.S. dollar surpassed 540 tenge at currency exchange offices on July 23 and continued to rise through July 25. The National Bank of Kazakhstan set the official exchange rate at 544.87 tenge per dollar and 639.2 tenge per euro for July 28.

Most Recent Stories

Kazakhstan Assesses Economic Out
Key Sectors Show Steady Expansion

Global Collaboration Key to Future o
Say IFLA Leaders During Visit to Kaz

President Tokayev to Pay Official Vis

Gravity, liquidity, bubbles, flows, pipes, stagnant/stagnation, drying up,

fluctuate - Latin *fluctuātus* undulated, past participle of *fluctuāre* to flow, equivalent. to *fluctu(s), a*; flowing (derivative of *fluere* to flow),

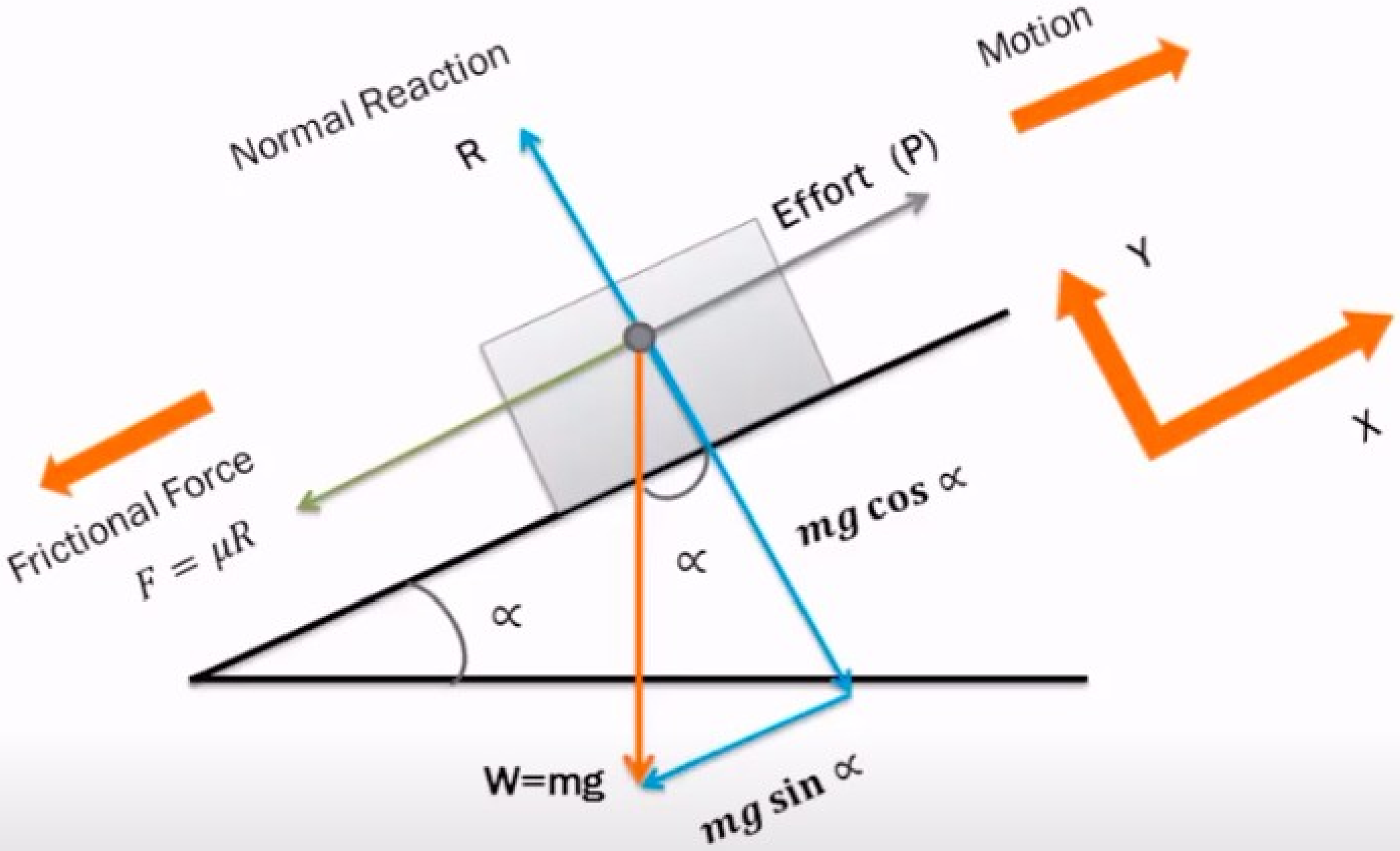
Inflation, floating exchange rate, market “forces”, “laws"

Last class

“economics concerns itself with the interplay of a variety of forces and that it is impossible to determine the resultant effect without calculus. For this reason, it is impossible to present economics without the aid of mathematics…

…a fact recognised in the case of pure astronomy, pure physics, mechanics and so forth”

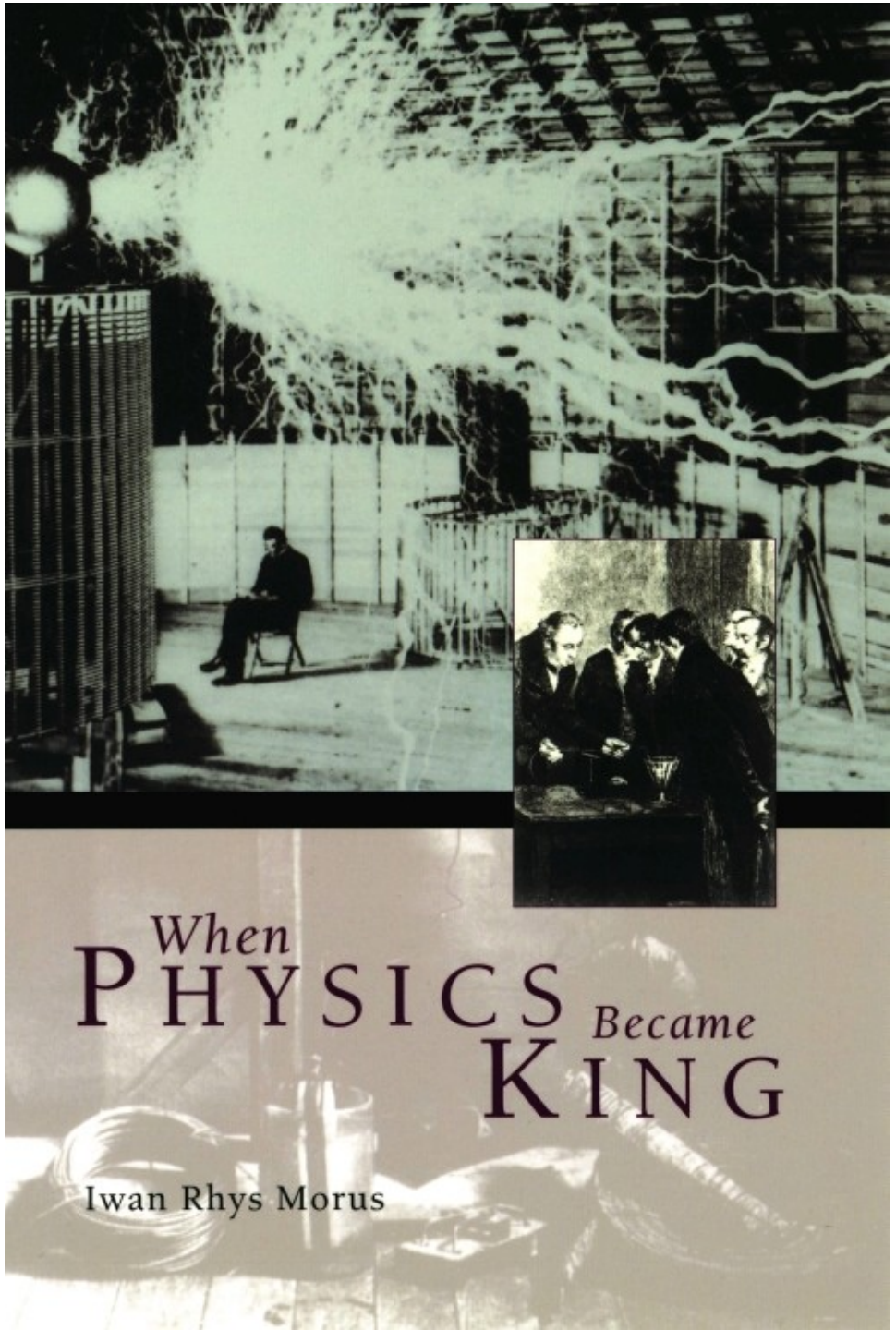
Prussian Economist Hermann Heinrich Gossen, 1854


Normal Reaction
R
Motion
Effort (P)
Y
X
Frictional Force
$F = \mu R$
$mg \cos \alpha$
α
α
W=mg
$mg \sin \alpha$

“(…) an active attempt to create a connection to the natural sciences was common among economists trying to make economics a rigorous science and was an important factor in promoting the use of mathematics. Creating an affinity between economics and natural sciences provided a justification to borrow concepts from natural sciences and make use of mathematical reasoning to “discover” general *laws* of economics. Portraying economics as a natural science was, furthermore, an attempt to endow economics with more authority. Almost all economists of the 19th century referred at some point to the natural sciences and their concepts when discussing economic questions.”

Nicolà Bezzola, *Mathematical Economics in the 19th Century*

Why Physics *specifically* ?



Light, wave theory, atoms, subatomic particles, electromagnetism, kinetic theory, thermodynamics, the ether, X rays…

a unifying theory of everything

1890s, Irving Fisher

THE PURCHASING POWER OF MONEY

ITS DETERMINATION
AND RELATION TO CREDIT
INTEREST AND CRISES

BY

IRVING FISHER

PROFESSOR OF POLITICAL ECONOMY IN YALE UNIVERSITY

ASSISTED BY

HARRY G. BROWN

INSTRUCTOR IN POLITICAL ECONOMY IN YALE UNIVERSITY

NEW AND REVISED EDITION

New York
THE MACMILLAN COMPANY
1920

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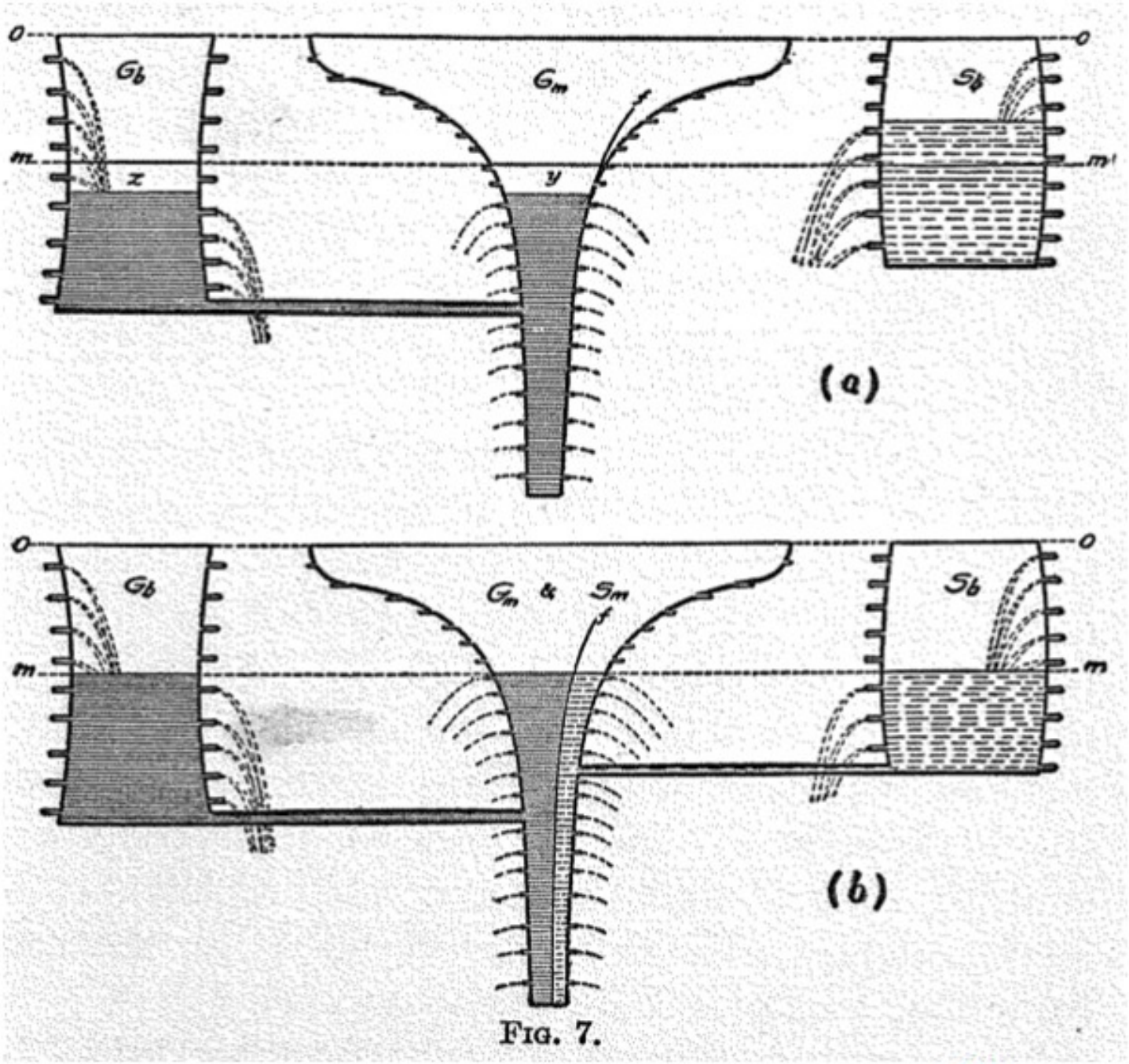
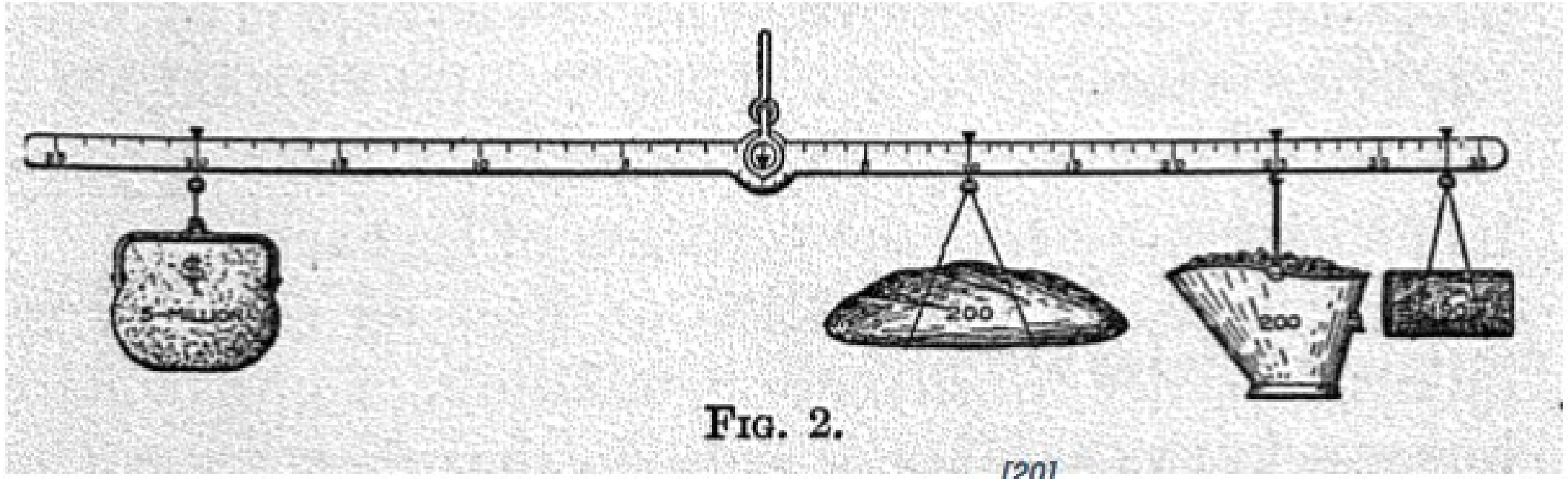


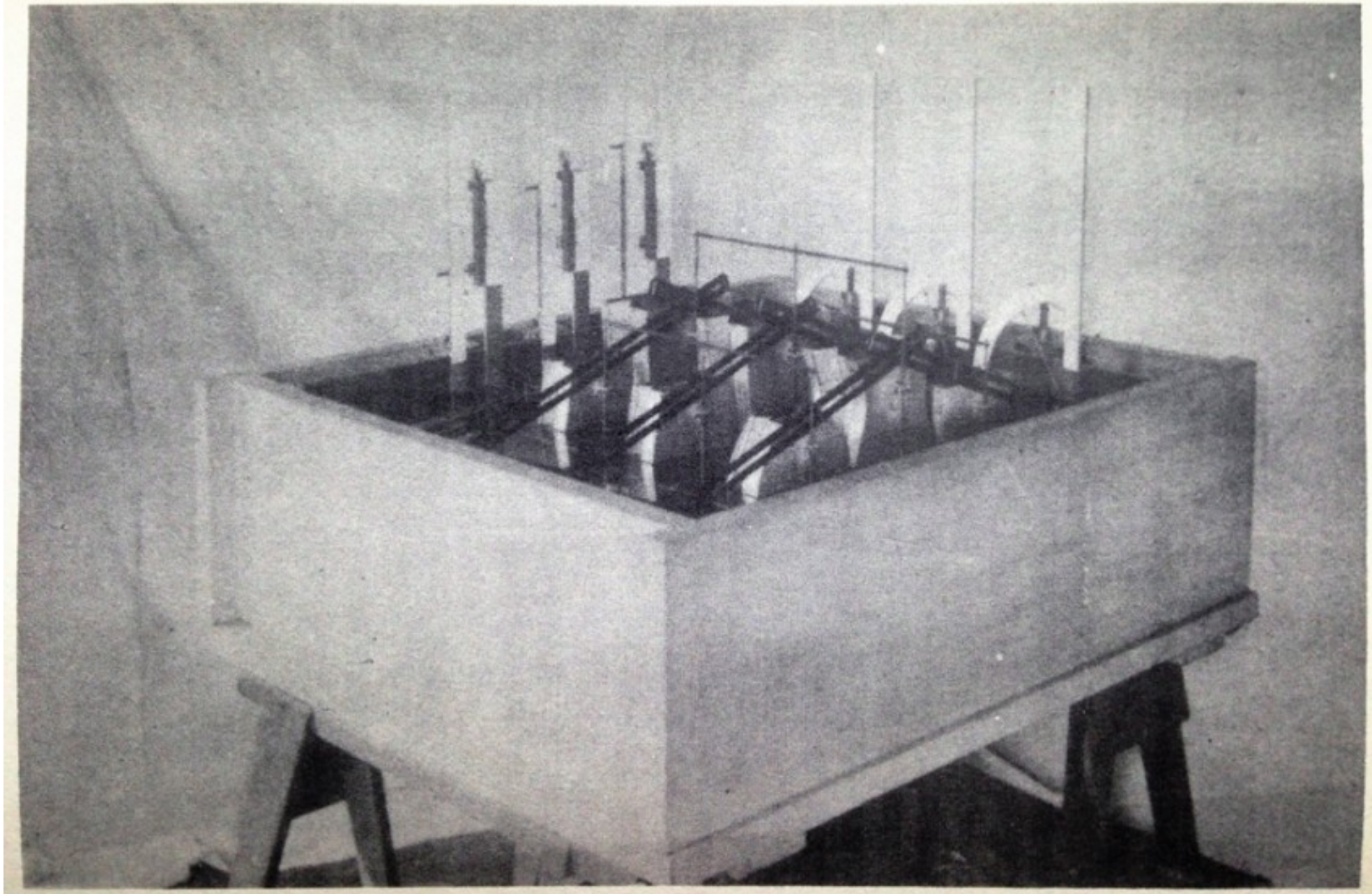
Figure 2: Fisher making use of hydraulics to model the purchasing power of money under a system of bimetallism [21]



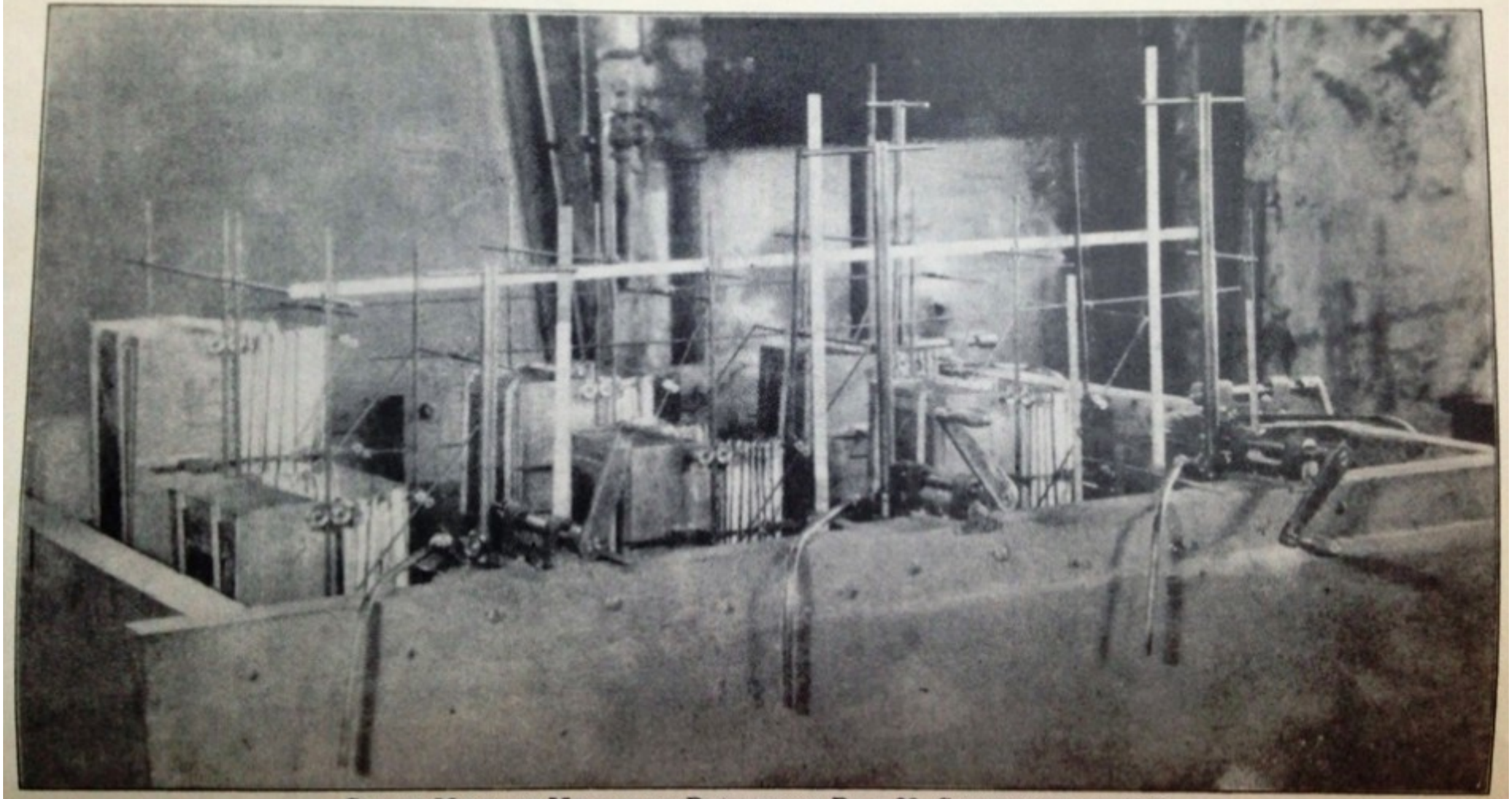
[20]

Figure 1: Fisher's mechanical balance of exchange to clarify the quantity theory of money.

Fisher 1891 PhD in mathematics and political economy (Yale’s first), studied general equilibrium analysis. He built a hydraulic mechanism to simulate the equilibrium prices and quantities—" a hydraulic computer in the days before electronic computers"

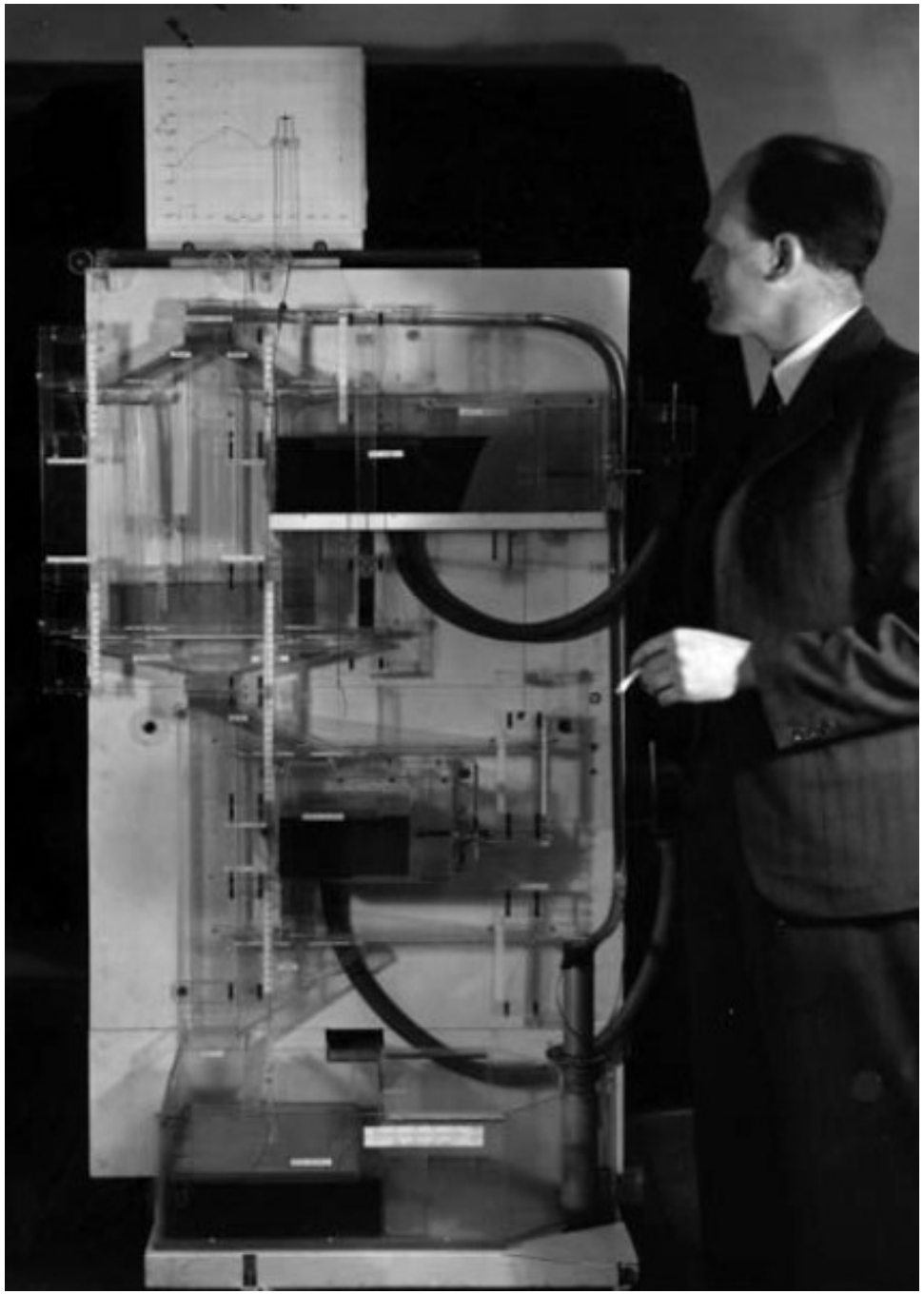


First Model of Mechanism Pictured on Page 38, Constructed in 1893.



Second Model of Mechanism Pictured on Page 38, Constructed in 1925.

1960s, Bill Phillips at LSE



MONIAC

Monetary National Income Analogue Computer

A hydraulic machine, made of transparent plastic pipes and tanks fastened to a wooden board, about six feet high, four feet wide, and three feet deep. The MONIAC represented the stocks and flows of a model and simulated how the model behaved as monetary and fiscal variables varied.

MONIAC

"Separate water tanks represent households, business, government, exporting and importing sectors of the economy. Coloured water pumped around the system measures income, spending and GDP. The system is programmable and capable of solving nine simultaneous equations in response to any change of the parameters, to reach a new equilibrium. It tracks changes in the trade balance, GDP and interest rates on paper. Simulation experiments with fiscal policy, monetary policy and exchange rates can be carried out: a teaching tool that can also forecast.

(a non-mathematical model – next class)

Federal Reserve, New Zealand

Break

Walras, 1870s

In “ a new branch of mathematics” Walras linked economics to Newton's laws.

Walras argued economics was a mathematical science, exploring the analogy between speed, space, and time, and scarcity, utility, and quantity, on the other.

Mirowski, *More Heat Than Light*, p. 224

Table 5.1 Fisher's Translations

Mechanics	Economics
*a particle	*an individual
space	commodity
force	marginal utility or disutility
*work	*disutility
energy	utility
*work or energy = force × space	*utility = marginal utility × commodity
force is a vector	marginal utility is a vector
forces are added by vector addition	marginal utilities are added by vector addition
work and energy are scalars	disutility and utility are scalars
The total energy may be defined as the integral with respect to impelling forces.	The total utility enjoyed by the individual is the like integral with respect to marginal utilities.
Equilibrium will be where net energy (energy minus work) is maximum; or equilibrium will be where impelling and resisting forces along each axis will be equal.	Equilibrium will be where gain (utility minus disutility) is maximum; or equilibrium will be where marginal utility and marginal disutility along each axis will be equal.
If total energy is subtracted from total work instead of vice versa the difference is "potential" and is a minimum.	If total utility is subtracted from total disutility instead of vice versa the difference may be called "loss" and is minimum.

Jevons

“(Utility) is difficult even to conceive; but it is the amount of utility which prompts us to buy, sell, consume, work, produce, and it is from the quantitative effects of those (salary, prices, etc) that we perceive utility…

… just like we can no more know nor measure gravity in itself, but we measure the motion of a pendulum”

Jevons

"in practice, no market ever long fulfils the theoretical conditions of equilibrium…

…but every branch of physics involves abstracting from the complexities of the actual world”

"The weight of a body is the sum of the weights of infinitely small particles, each acting at a different place, so that a mechanical problem resolves itself, strictly speaking, into an infinite number of problems. Equally, a price in a marketplace is the sum of infinitely scattered processes from all those individuals”

So we don’t need to know everything in the market to believe in The Market (class 3)

Were ever economics and physics *really* similar or have we imagined them and designed them such that now our economic intuition – our economic culture – cannot imagine itself out of it? Was this similarity willed into existence? Are we trapped in a 150 year long prestige anxiety trip?

Finance and Brownian motion, 1900s

THÉORIE

DE

LA SPÉCULATION,

PAR M. L. BACHELIER.

INTRODUCTION.

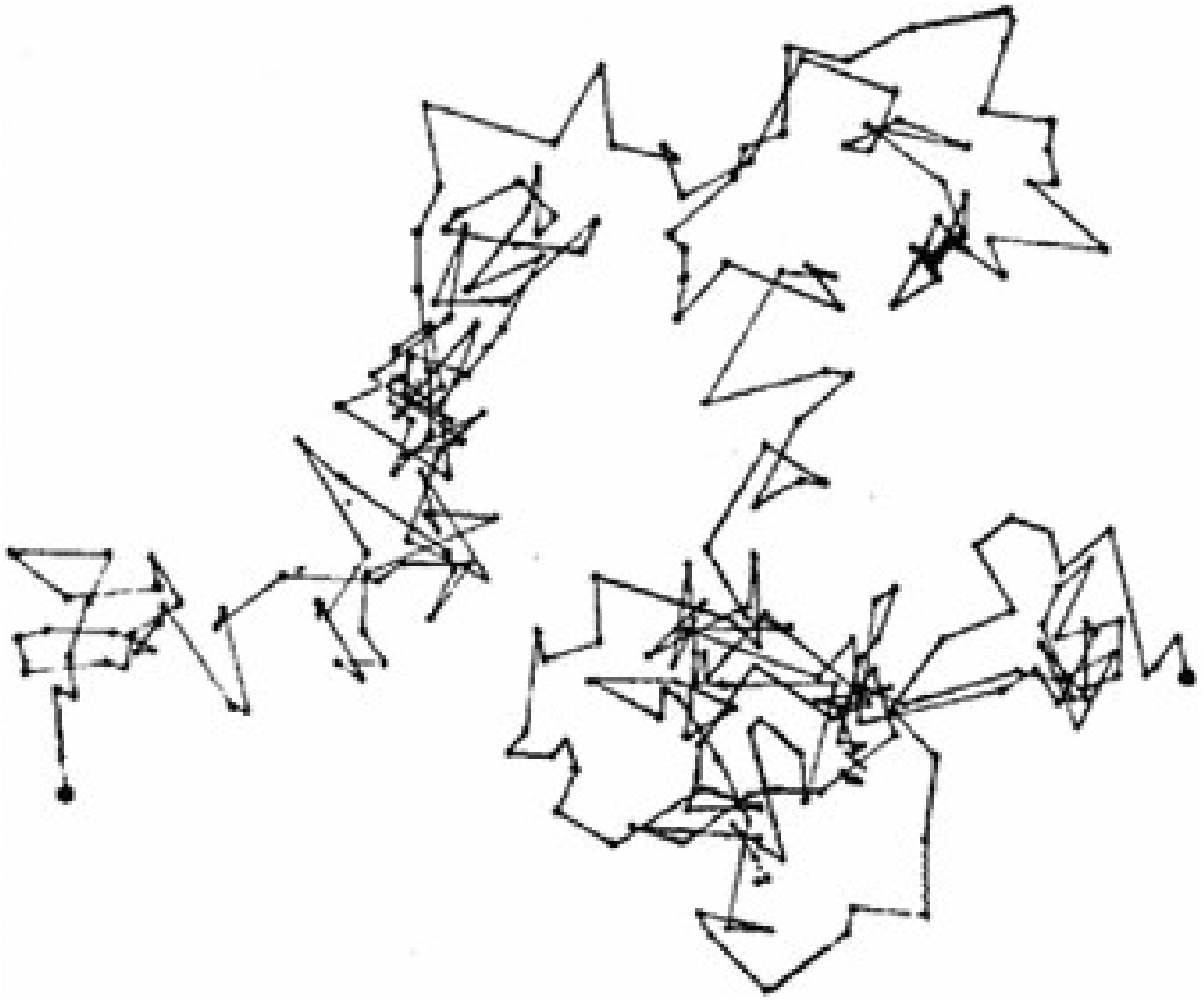
Les influences qui déterminent les mouvements de la Bourse sont innombrables, des événements passés, actuels ou même escomptables, ne présentant souvent aucun rapport apparent avec ses variations, se répercutent sur son cours.

A côté des causes en quelque sorte naturelles des variations, interviennent aussi des causes factices : la Bourse agit sur elle-même et le mouvement actuel est fonction, non seulement des mouvements antérieurs, mais aussi de la position de place.

La détermination de ces mouvements se subordonne à un nombre infini de facteurs : il est dès lors impossible d'en espérer la prévision mathématique. Les opinions contradictoires relatives à ces variations se partagent si bien qu'au même instant les acheteurs croient à la hausse et les vendeurs à la baisse.

Le Calcul des probabilités ne pourra sans doute jamais s'appliquer aux mouvements de la cote et la dynamique de la Bourse ne sera jamais une science exacte.

Mais il est possible d'étudier mathématiquement l'état statique du marché à un instant donné, c'est-à-dire d'établir la loi de probabilité des variations de cours qu'admet à cet instant le marché. Si le marché, en effet, ne prévoit pas les mouvements, il les considère comme étant



10142

THÉORIE

ANALYTIQUE

DE LA CHALEUR,

PAR M. FOURIER.

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A PARIS,

CHEZ FIRMIN DIDOT, PÈRE ET FILS,

LIBRAIRES POUR LES MATHÉMATIQUES, L'ARCHITECTURE HYDRAULIQUE ET LA MARINE, RUE JACOB, N° 24.

1822.

The heat equation

$$\alpha\left(\frac{\partial^2 T}{\partial x^2}+\frac{\partial^2 T}{\partial y^2}+\frac{\partial^2 T}{\partial z^2}\right)+\frac{\dot{q}_G}{\rho C_p}=\frac{\partial T}{\partial t}$$

Pricing Options with Fourier Series P1

Alexander Tsoskounoglou · Follow

6 min read · Jun 19, 2023

Option Pricing Formulae using Fourier Transform: Theory and Application

The COS Method: An Efficient Fourier Method for Pricing Financial Derivatives

Martin Schmelzle*

April 2010

Abstract

Why we care
First

Through these conscious, deliberate borrowings and transliterations, Economics took on the “seriousness” and the hardness of physics, its sense of inevitability and its authority

It’s less about whether “it’s right or wrong” and more about how we come to think about economic problems

Second

Because is Economics like Physics or is it LIKE Physics?

To oppose an argument that is not only perfect and beautiful (mathematics, classes 4 and 5) but also natural and, like natural phenomena, inevitable (physics, today), is not just stupid, ignorant or wrong – it’s against nature, morally scandalous, or in other words – Wrong.

Conclusion